

Project Design Phase

Solution Architecture

Date	20 JUNE 2026
Team ID	INDIVIDUAL PROJECT
Project Name	Online Complaint Registration and Management System
Maximum Marks	4 Marks

Solution Architecture:

Architectural Blueprint

Stack: Built on the MERN stack (MongoDB, Express.js, React.js, Node.js) to ensure scalability and modular development.

Frontend: React-based responsive interface designed for citizens, agents, and administrators, enabling seamless complaint registration and tracking.

Backend: Node.js with Express.js handles RESTful APIs for complaint submission, status updates, and secure authentication.

Database: MongoDB stores complaint records, user profiles, and resolution logs using structured Mongoose schemas for flexible data management.

Authentication: Implements JWT-based authentication for secure login and role-based access control (Admin, Agent, User).

Technical Performance

Real-Time Updates: WebSocket integration for instant complaint status notifications.

Optimized Queries: Indexed MongoDB collections to ensure fast retrieval of complaint data and user records.

Target Latency: Designed to process and return API responses within 250 ms for smooth user experience.

Security: Encrypted data transmission (HTTPS) and secure token validation to protect user information.

Development Roadmap

Phase 1: Database schema design, complaint model creation, and Express API setup.

Phase 2: Integration of JWT authentication, complaint assignment logic, and admin dashboard.

Phase 3: Real-time tracking module, notification system, and performance optimization for high user load.

Complaint Management System Architecture

